POLYMERIC MEMBRANES

A Comprehensive Learning Review

From fundamentals and phase separation to advanced hollow fiber CO_2 capture systems. Polymer selection · Dope formulation · Morphology control · Performance benchmarking · Current frontiers.

Coverage	**Structure · Phase inversion · Polymer chemistry · Dope solvents · Transport mechanisms · Modification strategies**
Key Polymers	PVDF · PAN · PES · PSf · PDMS · PI · PIM-1 · CA · Nafion · PEBA · Nylon · Cellulose
Applications	CO_2 capture (HFMC) · Gas separation · UF / MF / NF / RO · Ion exchange · Pervaporation
Figures	7 original scientific diagrams (classification · NIPS · morphology · Robeson · HFMC · transport)
Prepared for	Hamid — Polymeric Membrane Scientist · Calgary, Alberta · May 2026

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1. WHAT IS A MEMBRANE?

A membrane is a selective barrier — a phase (solid, liquid, or gas) that mediates the transport of one or more species between two adjacent phases while restricting others. In the context of polymer science and chemical engineering, membranes are thin films or tubes that exploit differences in size, charge, solubility, or diffusivity to achieve separation.

The membrane does not perform work itself — it exploits a driving force (pressure, concentration, electrical potential, temperature gradient) to selectively transport species. This makes membranes inherently energy-efficient compared to thermally driven separations such as distillation or evaporation.

KEY DEFINITION	A membrane is a condensed-phase region (typically 0.1 μm – 1 mm thick) that controls the relative rates of passage of different chemical species. Selectivity and permeability are the two fundamental, competing performance metrics.

1.1 Why Polymeric Membranes?

Polymers dominate membrane fabrication because they are processable: they dissolve in solvents, melt above their glass transition or melting temperature, and can be extruded, cast, or electrospun into complex geometries (flat sheets, hollow fibers, spiral-wound elements). Compared to inorganic alternatives (ceramic, zeolite, carbon molecular sieve), polymeric membranes offer:

- Lower capital cost — solution processing vs. high-temperature sintering
- Scalable manufacturing — continuous casting lines, dry-jet wet spinning
- Chemical tunability — copolymerization, blending, surface grafting, crosslinking
- Flexibility and self-supporting mechanical integrity in hollow fiber geometry
- Disadvantages: limited thermal stability (most polymers degrade above 300–400°C), physical aging (free volume collapse in glassy polymers), and plasticization by CO_2/hydrocarbons

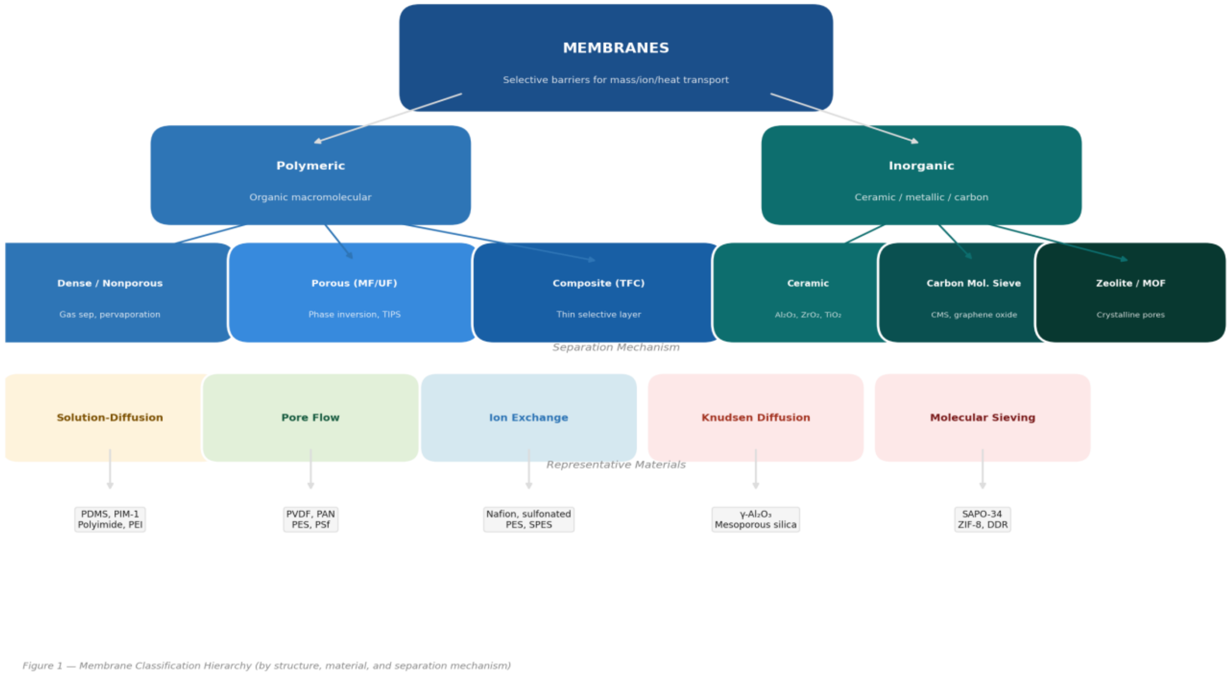

Figure 1 — Membrane classification hierarchy: by structure, material family, separation mechanism, and representative materials

1.2 Membrane Geometry

Two primary geometries dominate industrial and research applications:

Geometry	Description	Packing Density	Key Application	Fabrication Method
Flat Sheet	Planar film, supported on nonwoven backing	Low (100–400 m^2/m^3)	RO, NF, UF laboratory	Doctor blade, slot-die coating
Hollow Fiber	Self-supporting tube, lumen inside	Very high (500–9000 m^2/m^3)	Gas separation, CO_2/amine HFMC, hemodialysis	Dry-jet wet spinning
Spiral-Wound	Flat sheet wrapped around central tube	High (300–1000 m^2/m^3)	RO, NF water treatment	Module assembly
Tubular	Large-bore tube (ID > 3 mm)	Low (20–200 m^2/m^3)	Harsh-feed MF, industrial bioprocess	Extrusion or casting on tube support

2. PHASE INVERSION — HOW MEMBRANES ARE MADE

The dominant method for fabricating polymeric membranes is phase inversion — a thermodynamically controlled process in which a homogeneous polymer solution is transformed into a two-phase system (solid polymer-rich + liquid polymer-lean), generating the porous structure. There are four principal variants.

2.1 Non-Solvent Induced Phase Separation (NIPS)

NIPS (also called immersion precipitation) is the cornerstone process for UF, MF, and hollow fiber membranes. A polymer dope (polymer + solvent ± additives) is cast or extruded, then immersed in a non-solvent bath (typically water). Rapid solvent–non-solvent exchange drives the system across the binodal into the two-phase region of the ternary phase diagram.

MECHANISM	The exchange kinetics determine morphology: fast exchange (high affinity solvent/non-solvent pair, e.g., NMP/water) → finger-like macrovoids below a dense skin. Slow exchange (lower affinity, or with co-solvent in bath) → symmetric sponge-like structure with higher mechanical strength but lower flux.

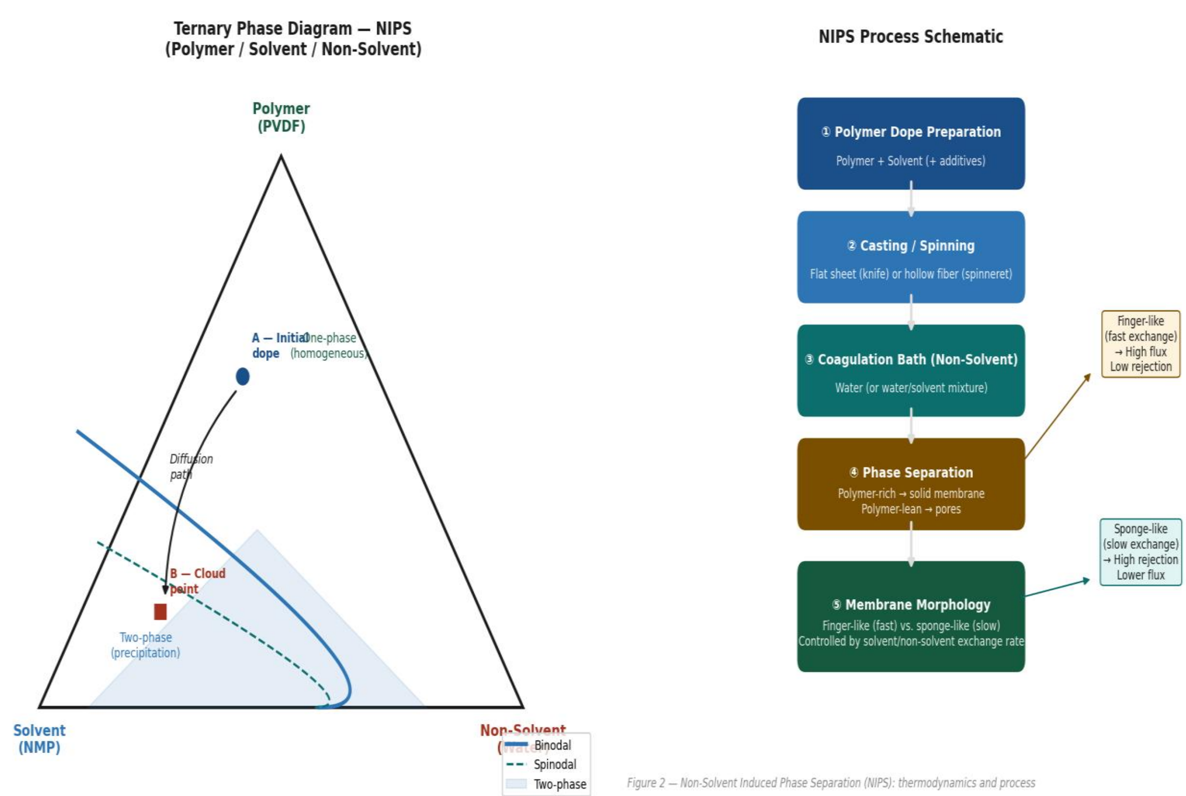

Figure 2 — NIPS thermodynamics (ternary phase diagram) and process schematic: from dope preparation to membrane morphology

2.2 The Ternary Phase Diagram — Critical Concepts

Understanding the ternary (polymer / solvent / non-solvent) phase diagram is the fundamental tool for rationalizing NIPS membrane morphology:

Region / Feature	Definition	Membrane Consequence
Homogeneous region (one-phase)	System miscible — above binodal	The starting dope must sit here to be castable
Binodal curve	Boundary of thermodynamic stability — two phases coexist	Crossing binodal initiates phase separation
Spinodal curve	Limit of metastability — spontaneous spinodal decomposition	Inside spinodal: bicontinuous interconnected pore structure
Critical point	Apex of spinodal inside binodal	Determines co-continuous vs. droplet morphology
Gelation line	Polymer-rich phase solidifies (vitrification or crystallization)	Sets final pore dimensions — below this, structure is locked
Tie-line	Connects compositions of two coexisting phases at equilibrium	Determines pore-wall polymer concentration and skin density
Diffusion path A→B	Trajectory of local composition during non-solvent ingress	Slope of path relative to binodal determines nucleation mode

2.3 Thermally Induced Phase Separation (TIPS)

In TIPS, the polymer is dissolved in a high-boiling diluent (solvent) at elevated temperature. On cooling, the solution demixes due to decreased polymer solubility. TIPS is preferred for semi-crystalline polymers (PVDF, PP, PE) that are difficult to dissolve at room temperature and for which NIPS gives insufficient mechanical strength.

Parameter	NIPS	TIPS
Driving force	Solvent–non-solvent exchange (mass transfer)	Temperature decrease (thermodynamic)
Polymer types	Amorphous / glassy (PES, PSf, PAN, CA)	Semi-crystalline (PVDF, PP, PE, PVDF-HFP)
Diluent	Good solvent (NMP, DMAc, DMF)	Poor solvent at RT (glycerol, sebacic acid, DBP, mineral oil)

Pore structure	Asymmetric (finger + skin)	Symmetric sponge or cellular; highly interconnected
Pore size control	Polymer conc., additive, bath composition	Cooling rate, diluent composition, crystallization nuclei
Key advantage	Thin skin → high flux asymmetric UF/NF membranes	PVDF fibers with high mechanical strength; MF membranes
Key limitation	Solvent recovery required; skin defects from vibration	Diluent extraction step adds complexity; fewer commercial solvents

2.4 Dry-Phase Separation (VIPS) and Electrospinning

Vapor-Induced Phase Separation (VIPS): the cast film is exposed to a humid atmosphere (water vapor) rather than immersed in liquid. Water vapor condenses selectively on the surface, inducing nucleation of droplet-type pores. Results in a more open-cell morphology than NIPS. Used for PVDF membranes with high hydrophobicity.

Electrospinning: a high-voltage electric field draws a polymer jet from a needle onto a collector, producing a nanofibrous non-woven mat (fiber diameter 50 nm – 5 µm). Pores are the interfiber spaces (not within fibers). Used for nanofiber filtration membranes, tissue scaffolds, and as a substrate for TFC membranes.

3. MEMBRANE MORPHOLOGY AND STRUCTURE

Membrane morphology describes the three-dimensional arrangement of polymer and void (pore) space. It directly controls permeability (flux) and selectivity (rejection or separation factor). Four canonical morphologies are recognized:

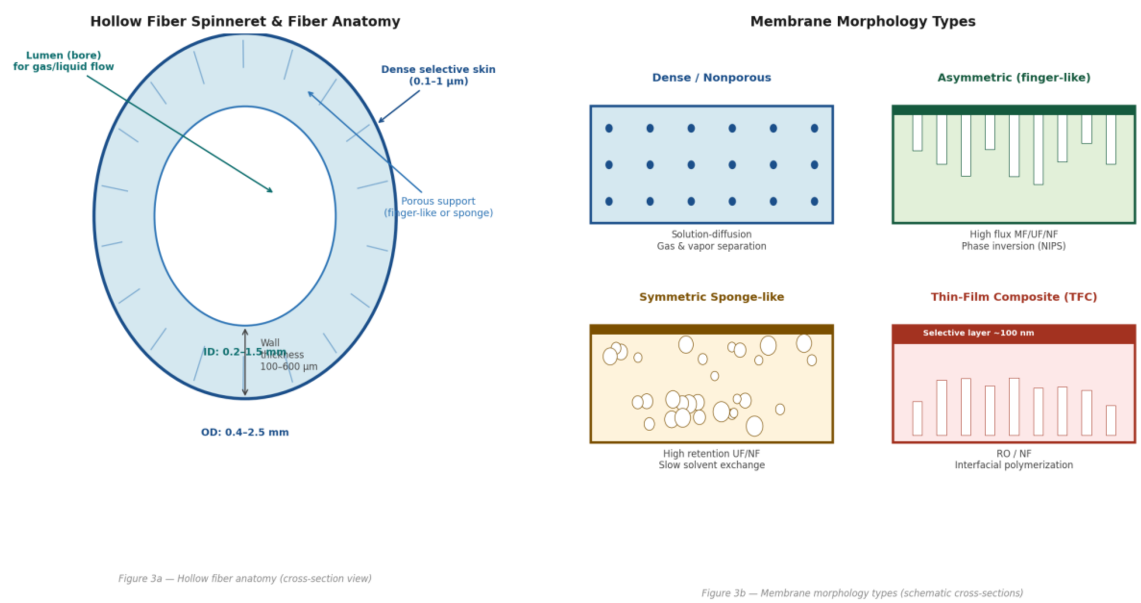

Figure 3 — Membrane morphology types: hollow fiber anatomy (cross-section) and four canonical structures

3.1 Dense (Nonporous) Membranes

No continuous pore channels. Transport occurs entirely by solution-diffusion: molecules dissolve at the high-activity (high-pressure) face, diffuse through the polymer matrix, and desorb at the low-activity face. Selectivity determined by the relative solubility (S) and diffusivity (D) of each species in the polymer: $P = S \cdot D$.

- Materials: PDMS, polyimide (6FDA-series), PEBA, PIM-1, PTMSP, CA
- Applications: gas separation (CO_2/CH_4, O_2/N_2, H_2/N_2), vapor permeation, pervaporation
- Typical thickness: 1 µm – 100 µm (dense selective layer in TFC: 50–200 nm)

3.2 Asymmetric Membranes (Finger-like Macrovoids)

A thin, dense skin (0.1–1 µm) formed during the very early moments of NIPS sits atop a thick spongy or finger-like substructure. The skin provides selectivity; the substructure provides mechanical support with minimal hydraulic resistance. The Loeb-Sourirajan membrane (1963) was the first asymmetric CA membrane — a historic milestone that enabled practical reverse osmosis.

- Skin formation: rapid polymer precipitation at the film/bath interface → dense, defect-minimized polymer-rich layer
- Macrovoid formation: differential exchange rate creates a polymer-lean zone that nucleates large voids (finger-like channels)
- Control parameters: polymer MW, dope concentration, air gap (hollow fiber), bath temperature, additive (PVP, LiCl, water in dope)

3.3 Symmetric Sponge-like Membranes

Slow solvent–non-solvent exchange (achieved by adding solvent to the bath, lowering bath temperature, or using a low-affinity non-solvent) gives a uniform sponge-like structure with narrow pore size distribution. Higher mechanical strength than finger-like morphology. Used for MF membranes (pore diameter 0.1–10 μm) where flux is less critical than robustness.

3.4 Thin-Film Composite (TFC) Membranes

A TFC membrane consists of three layers: (i) a non-woven polyester support (mechanical backbone, ~150 μm); (ii) a microporous polymeric support membrane (PSf or PAN, ~50 μm); and (iii) an ultra-thin selective polyamide layer (~50–200 nm) formed by interfacial polymerization (IP) between a diamine (MPD, dissolved in water) and an acid chloride (TMC, dissolved in hexane) at the water–organic interface.

- Achievable flux: >40 L/m^2h at 15 bar (seawater RO)
- Salt rejection: >99.7% NaCl at 55–60 bar (SWRO)
- Active layer chemistry: fully aromatic cross-linked polyamide (m-phenylenediamine + trimesoyl chloride)
- Commercial examples: Dow FilmTec BW30, TORAY SU-720, Nitto ESPA

4. MEMBRANE TRANSPORT MECHANISMS

The mechanism by which species traverse a membrane is determined by the membrane's physical structure, the nature of the permeating species, and the driving force. Selecting the wrong membrane type for an application leads to either poor selectivity or unacceptably low flux.

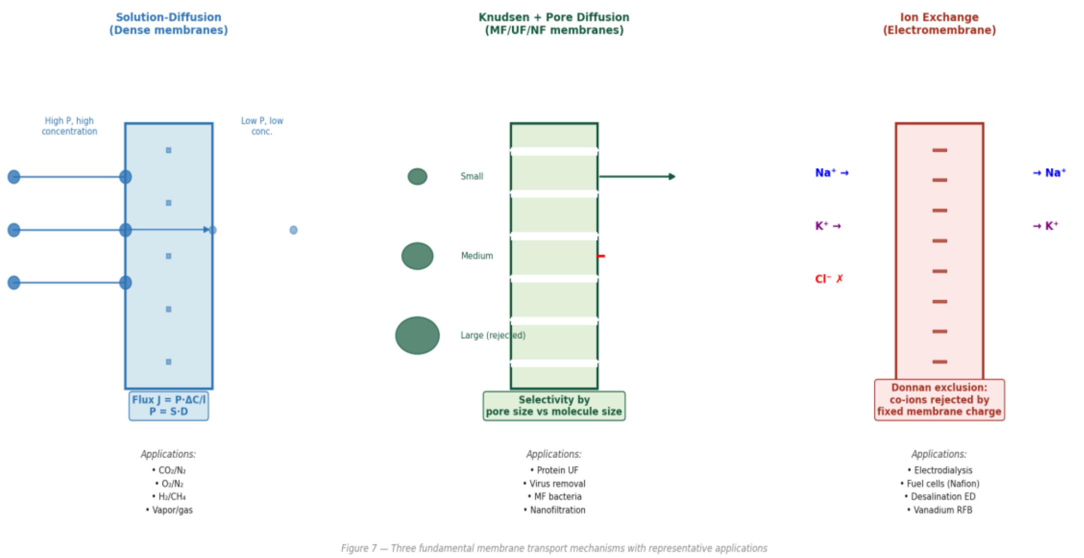

Figure 4 — Three fundamental transport mechanisms: solution-diffusion (dense), pore flow/Knudsen (MF/UF), and ion exchange (electromembrane)

4.1 Solution-Diffusion Model

The dominant model for dense polymeric membranes (gas separation, pervaporation, RO). Assumptions: (1) the membrane is defect-free and nonporous; (2) transport is at steady state; (3) chemical potential gradient is the driving force throughout the membrane thickness l.

FLUX EQUATION	$J = P \cdot \Delta C / l = (S \cdot D) \cdot \Delta C / l$ where P = permeability (Barrers), S = solubility coefficient ($cm^3(STP)/cm^3 \cdot cmHg$), D = diffusion coefficient (cm^2/s), l = membrane thickness, ΔC = concentration driving force across membrane.

Selectivity (ideal, or permselectivity): $\alpha(A/B) = PA/PB = (SA/SB) \cdot (DA/DB)$. For CO_2/N_2: CO_2 is both more soluble (S ratio ~10–20) and more diffusible in most rubbers and some glassy polymers. CO_2 has a kinetic diameter of 3.30 Å vs. N_2 at 3.64 Å — a small but exploitable difference in stiff glassy polymer matrices.

4.2 Pore Flow and Knudsen Diffusion

In porous membranes (MF, UF, and the porous sublayer of NF), flow occurs through continuous pore channels. The appropriate model depends on the ratio of pore diameter dp to the mean free path of the gas molecule λ:

Flow Regime	Criterion	Rate-Controlling Step	Selectivity Basis
Viscous (Poiseuille)	dp >> λ (dp > 100 nm for gases)	Pressure-driven bulk flow	None (no separation)
Knudsen diffusion	dp < λ (dp ~ 2–100 nm for gases at 1 atm)	Molecule–wall collisions dominate	$\alpha = \sqrt{M_B/M_A}$ — light gases faster
Surface diffusion	Strong adsorption on pore walls	Adsorbed phase diffusion along wall	Adsorptive selectivity
Molecular sieving	dp ≈ molecule size (< 1 nm)	Steric exclusion at pore entrance	Size-based, ultra-sharp cutoff
Liquid filtration (UF)	dp > 1 nm, liquid feed	Hydraulic resistance (Hagen-Poiseuille)	Molecular weight cutoff (MWCO)

4.3 Ion Exchange Transport (Electromembranes)

Ion exchange membranes carry fixed charged groups covalently attached to the polymer backbone. Cation exchange membranes (CEM) carry negative fixed charges (sulfonate $–SO_3^-$, carboxylate $–COO^-$) and pass cations while excluding anions (Donnan exclusion). The opposite holds for anion exchange membranes (AEM).

- Nafion® (DuPont): sulfonated PTFE perfluoropolymer; proton conductivity ~0.1 S/cm; gold standard for PEM fuel cells and electrolyzers
- Electrodialysis: alternating CEM/AEM stack with electrical driving force — produces desalted stream and brine from seawater
- Vanadium redox flow batteries: Nafion membrane separates V^{2+}/V^{3+} from V^{4+}/V^{5+} while conducting H^+ ions
- Bipolar membranes: CEM/AEM laminate that drives water splitting into H^+ and OH^- under reverse bias

5. KEY POLYMERIC MEMBRANE MATERIALS — DETAILED PROFILES

This section provides detailed profiles of the twelve most important polymers in membrane science, covering structure, properties, dope formulation, phase-inversion conditions, morphology tendencies, and primary applications.

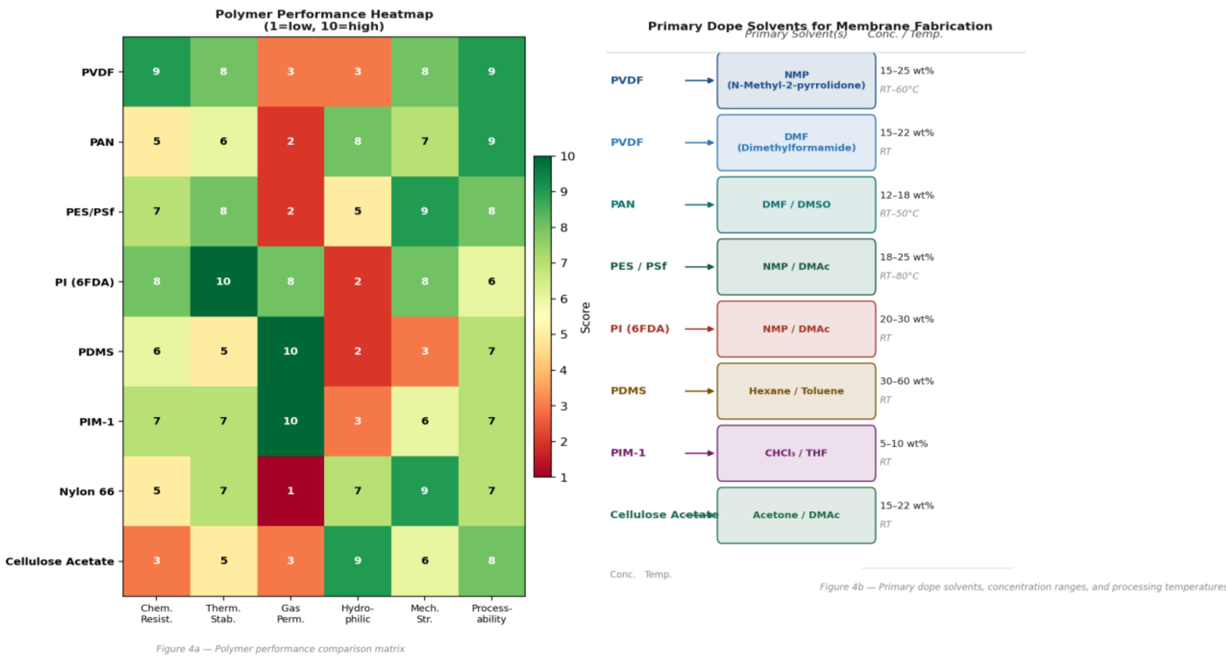

Figure 5 — Polymer performance heatmap and primary dope solvent guide for membrane fabrication

5.1 PVDF — Polyvinylidene Fluoride — Semi-crystalline fluoropolymer · Workhorse of MF/UF and HFMC

PVDF ($–CH_2–CF_2–)_n$ has a molecular weight of 200–700 kDa. It is semi-crystalline (α, β, γ phases; β-phase piezoelectric). Outstanding chemical resistance to most acids, bases (pH 2–13), and organic solvents. Hydrophobic surface (contact angle 80–90°) — critical for anti-wetting performance in direct-contact membrane contactors (DCMD, HFMC for CO_2 capture). Processed via NIPS (NMP or DMF) and TIPS (glycerol, triethylene glycol, DMSO/water).

Backbone	$–(CH_2–CF_2)_n–$ | MW: 200–700 kDa | Crystallinity: 35–70%
Tg / Tm	Tg: −40°C | Tm: 155–175°C | Decomposition: >450°C
Primary solvents	NMP (best: 15–25 wt%, 60°C) · DMF (15–22 wt%, RT) · DMAc · DMSO/acetone blends
TIPS diluents	Glycerol (high Tm), triethylene glycol (TEG), sebacic acid, DMSO/H_2O 70:30
Additives in dope	LiCl (1–3 wt%, pore former) · PVP (macropore suppressor) · SiO_2 NPs · water (co-non-solvent)

Typical dope	PVDF 18%, NMP 76%, LiCl 3%, H_2O 3% · viscosity ~5,000–15,000 cP at 25°C
Coagulation bath	Water (0–60°C) · NMP/water for sponge morphology · air gap: 0–20 cm (hollow fiber)
Morphology tendency	Finger-like macrovoids (NMP/water, fast exchange) · Sponge + small pores (TIPS, or NMP/water+30% NMP in bath)
Contact angle	Pristine: 80–90° · After SiO_2-HDTMS modification: >150° (superhydrophobic)
Key limitation	Dehydrofluorination above pH 13 (base attack); physical aging minimal (semi-crystalline)
CO_2/amine HFMC	Best substrate for K-glycinate, K-argininate systems — high LEP critical; SI-ATRP of HDTMS recommended

Membrane applications: MF/UF water treatment · Direct-contact membrane distillation (DCMD) · CO_2/amine HFMC · Piezoelectric sensor membranes · Hollow fiber blood oxygenators · Li-ion battery separators

5.2 PAN — Polyacrylonitrile — Glassy, hydrophilic UF workhorse with excellent chemical stability

PAN (–CH_2–CH(CN)–$)_n$, MW 70–150 kDa. The nitrile group (–CN) imparts high polarity, partial hydrophilicity, and strong hydrogen-bond acceptor capacity. PAN is a carbon fiber precursor (cyclization and carbonization at 250–1400°C → CMS membranes). Excellent solvent resistance to hydrocarbons. Forms asymmetric UF/NF membranes with MWCO 1–150 kDa by NIPS. The standard substrate for TFC membranes when good adhesion to interfacial polyamide layer is needed.

Backbone	–(CH_2–CH(CN)$)_n$– | MW: 70–150 kDa | Amorphous–semicrystalline
Tg / Tm	Tg: 85–105°C | No clean Tm (degrades before melting in pure form)
Primary solvents	DMF (12–18 wt%, RT–50°C) · DMSO (12–18 wt%, 40–60°C) · NMP · PC (propylene carbonate)
Typical dope	PAN 14%, DMF 83%, LiCl 3% · or PAN 15%, DMSO 85% · viscosity: 2,000–8,000 cP
Coagulation bath	Water (20–60°C) · H_2O/DMF (10–30% DMF for sponge structure)
Morphology tendency	Strongly finger-like (DMF/H_2O, fast diffusivity) · Dense skin + macrovoids below
Surface character	Mildly hydrophilic (contact angle ~50–65°) · –CN groups can be hydrolyzed to –COOH (PAN-g-COOH)

Key advantage	Low cost, available in broad MW range; excellent as TFC NF/RO substrate; easily functionalized
Key limitation	Degrades in strong alkalis (saponification of CN); limited thermal stability above 200°C
Surface mod	Alkaline hydrolysis (NaOH, 1M, 60°C, 30 min) → –COOH groups for covalent grafting of polyamide TFC layer

Membrane applications: UF protein/enzyme separation · TFC substrate for NF/RO · Carbon molecular sieve membrane precursor · pH-responsive NF (DMAEMA grafted) · Blood dialysis membranes (low endotoxin PAN)

5.3 PES / PSf — Polyethersulfone / Polysulfone — Thermally stable glassy polymer · Standard UF/MF substrate · Excellent film former

Polysulfone (PSf, Udel® MW ~35 kDa) and polyethersulfone (PES, Ultrason® E MW ~50–70 kDa) share an aryl–SO_2–aryl backbone conferring extraordinary hydrolytic and thermal stability (Tg 185–230°C). The slightly more hydrophilic PES is preferred for biomedical UF. Both form excellent asymmetric membranes via NIPS in NMP or DMAc. The standard membrane substrate in hemodialysis and plasma fractionation modules. Solvent resistance: good to water, alcohols, dilute acids; attacked by ketones, chlorinated solvents, and strong DMF at elevated temperatures.

Backbone	PSf: bisphenol A + diphenyl sulfone | PES: –C_6H_4–SO_2–C_6H_4–O– | MW: 35–80 kDa
Tg / Tm	PSf Tg: 185°C | PES Tg: 220–230°C | Amorphous — no Tm
Primary solvents	NMP (18–25 wt%, RT–80°C) · DMAc (20–26 wt%) · DMF · NMP/acetone blends
Typical dope	PES 20%, NMP 76%, PVP K30 4% · viscosity: 8,000–25,000 cP · PVP improves hydrophilicity
Coagulation bath	Water (20–60°C) · Water/NMP (10–20% NMP for sponge sublayer) · EtOH/water (hollow fiber bore fluid)
Morphology tendency	Thin dense skin + thick finger-like sublayer (NMP/water) · MWCO tunable 1–150 kDa by conc. and additives
Key additives	PVP K30/K90 (hydrophilization, macrovoid suppressor) · PEG 400–6000 (pore former) · LiCl · glycerol (TIPS adjuvant)
Contact angle	~65–80° (hydrophobic) · PVP addition: ~50–65° · Grafting: < 30° (hydrophilic brushes)

Key application highlight	PSf or PES hollow fibers are the substrate for most commercial TFC NF membranes and hollow fiber blood dialyzers worldwide
Limitation	Poor resistance to ketones (MEK, acetone) and chlorinated solvents; UV degradation; hydrophobic fouling by proteins

Membrane applications: Hemodialysis (hemofilter) · TFC NF/RO support · Plasma fractionation · UF for dairy (whey protein) · Gas separation substrate · Biomedical implant membranes

5.4 Polyimide (PI) — 6FDA-based and Matrimid — Thermally robust glassy polymer · Premier gas separation material

Polyimides are condensation polymers of a dianhydride and diamine, forming a cyclic imide linkage. 6FDA-based PIs (hexafluorodianhydride + bulky diamine such as DAM, DABA, TMPDA, or TMBP) have exceptional CO_2 permeability (10–2000 Barrers) and CO_2/CH_4 selectivity (30–50) — near the Robeson upper bound. Matrimid® 5218 (BTDA + DAPI) is the commercial benchmark. All PIs are glassy at RT and exhibit physical aging and CO_2-induced plasticization — the two dominant failure modes in biogas upgrading membranes.

Backbone	–[Ar–CO–N(R)–CO–Ar–O–Ar]– imide ring linkage | MW: 50–200 kDa | Amorphous glassy
Tg	Matrimid: 305°C | 6FDA-DAM: 365°C | 6FDA-TMPDA: 350°C | Stable to ~450°C in inert atmosphere
Primary solvents	NMP (20–30 wt%, RT) · DMAc · THF/dioxane blends · NMP/THF for lower viscosity
Typical dope	Matrimid 25%, NMP 75% · viscosity: 15,000–50,000 cP · or 6FDA-PI 20%, DMAc 80%
Coagulation bath	Water (20–40°C) · MeOH for flat-sheet dense films · Air evaporation for dense dense membranes
Dense film casting	Cast from 3–8 wt% solution in $CHCl_3$, evaporate slowly → dense defect-free film for gas permeation testing
Crosslinking	Thermal (350–450°C in N_2) → thermally rearranged (TR) PI → β-TR polymer with enhanced CO_2/N_2 > 50
Physical aging	Permeability drops 30–60% in 1–2 years; crosslinking or blending with PIM reduces aging rate

Plasticization	CO_2 at >15 bar plasticizes glassy PI → flux increases but selectivity collapses; crosslinking mitigates
Limitation	Expensive synthesis; NMP-intensive processing; brittle hollow fibers unless blended or copolymerized

Membrane applications: Natural gas sweetening (CO_2/CH_4) · Hydrogen purification (H_2/N_2) · O_2 enrichment · Biogas upgrading · CO_2 capture from flue gas · High-T filtration

5.5 PDMS — Polydimethylsiloxane — Rubbery silicone · Highest gas permeability of any polymer

PDMS (–Si(CH_3)$_2$–O–)$_n$, MW 50–200 kDa by anionic ROP of hexamethylcyclotrisiloxane (D3), has the lowest glass transition temperature (−127°C) of any polymer — far below room temperature — making it an ultra-rubbery elastomer at all practical conditions. Chain flexibility from the Si–O–Si backbone (bond angle 142°, rotation barrier near zero) gives extremely high free volume and gas diffusivity. CO_2 permeability > 3000 Barrers. Used as the selective rubbery layer in composite membranes for CO_2 removal from natural gas at low pressures.

Backbone	–(Si(CH_3)$_2$–O)$_n$– | MW: 50–200 kDa | Rubbery elastomer at RT
Tg / Crosslinking	Tg: −127°C | Must be crosslinked for membrane use (Pt-catalyzed hydrosilylation or peroxide)
Primary solvents	Hexane (30–60 wt%) · Toluene · heptane · Iso-octane — all non-polar
Film preparation	Crosslinked: PDMS + PMHS (crosslinker) + Pt catalyst in toluene → coat on porous support → cure 100°C, 1 h
Selective layer thickness	5–50 μm (dip-coated on PAN or PES support) · 100 nm–2 μm (spin-coated for gas permeation testing)
Key properties	CO_2 perm: 3000–5000 Barrers | CO_2/N_2 selectivity: 8–12 | O_2/N_2: ~2.2 | H_2O/N_2: very high
CO_2 selectivity basis	Solubility-driven: CO_2 much more soluble in rubbery PDMS than N_2 (S_CO_2/S_N_2 ~ 12)
Limitation	Low selectivity for gas separation; swells in hydrocarbons; difficult to make defect-free thin films alone
CO_2 application	Rubbery PDMS composite for CO_2 permeation at low partial pressure differentials; pervaporation of organics from water

Membrane applications: CO_2 removal from CH_4 (low selectivity, high permeability) · Organic solvent pervaporation · Dehumidification · Oxygen enrichment (medical) · Blood oxygenator coating · Anti-fouling coating for UF membranes

5.6 PIM-1 — Polymer of Intrinsic Microporosity — Rigid, contorted glassy polymer · Surpasses Robeson upper bound

PIM-1 (McKeown & Budd, 2004) is formed by double SNAr (nucleophilic aromatic substitution) condensation of 5,5',6,6'-tetrahydroxy-3,3,3',3'-tetramethyl-1,1'-spirobisindane (SBI) with tetrafluoroterephthalonitrile (TFTPN) under K_2CO_3/DMAc at 65°C. The spiro-carbon center forces a 90° contortion at every repeat unit, preventing chain packing and generating ~750 m²/g BET surface area from interconnected micropores (<2 nm). CO_2 permeability ~2000–5000 Barrers. Physical aging (free volume collapse) is the dominant long-term limitation.

Synthesis	SBI + TFTPN, K_2CO_3, DMAc, 65°C, 72 h → dibenzodioxin linkage; MW 50–200 kDa by GPC
BET surface area	700–850 m²/g (PIM-1) · 1000–1100 m²/g (triptycene-PIM) · Micropores < 2 nm
Primary solvents	$CHCl_3$ (5–8 wt%, RT) — excellent · THF (5–10 wt%) · DCM · Toluene (partial)
Membrane casting	Dissolve in $CHCl_3$, filter, cast on glass, slow evaporation (24–48 h, 20°C) → dense glassy film 20–100 μm
CO_2 permeability	2000–5000 Barrers (fresh) · 800–1500 Barrers after 1 year aging — physical aging is dominant
Physical aging mitigation	UV crosslinking of CN groups (tetrazole at 180°C, UV 254 nm) · Crosslinking reduces aging but reduces permeability
Tg	No observable Tg by DSC (rigid backbone prevents segmental motion at accessible temperatures)
Mixed-matrix membrane	PIM-1 particles (2–20 wt%) in PVDF or Matrimid matrix → synergistic CO_2/N_2 performance
Limitation	Physical aging: 40–60% permeability drop over 2 years at RT; brittleness; no hollow fiber capability alone

Membrane applications: CO_2/N_2 natural gas and flue gas separation · H_2 purification · O_2 enrichment · Gas mixed-matrix membranes (PVDF+PIM-1) · Pervaporation of organic mixtures · Nanofiltration of dyes

5.7 Cellulose Acetate (CA) — Classic hydrophilic polymer · First commercial RO membrane material

Cellulose acetate (CA), the first synthetic membrane polymer for desalination (Loeb and Sourirajan, 1963), is produced by acetylation of cellulose. Degree of substitution (DS) controls properties: CA-2.5 (DS = 2.5) for RO; CA-1.8 for UF. Naturally hydrophilic (abundant –OH and –OAc groups), excellent biocompatibility, and low protein adsorption. Limitation: narrow pH window (4–8), temperature sensitivity (hydrolysis > 35°C), and susceptibility to biological degradation. Now largely replaced by polyamide TFC for RO, but still dominant in gas separation (CO_2/CH_4 in natural gas plants) and hemodialysis (low complement activation).

Structure	Cellulose backbone with acetyl groups (DS 1.8–2.5); MW 30–150 kDa; semi-crystalline
Primary solvents	Acetone + formamide (4:1 v/v) · DMAc · Dioxane/methanol · NMP + $MgClO_4$
Typical dope	CA 22%, acetone 66%, formamide 10%, $MgClO_4$ 2% · or CA 20%, DMAc 80% (UF fibers)
Coagulation bath	Water/isopropanol (for RO asymmetric flat sheet) · Water 0–5°C (hollow fiber UF)
CO_2 permeability	~6–15 Barrers · CO_2/CH_4 selectivity: 15–22 · Used in spiral-wound modules for natural gas
Contact angle	~50–60° (hydrophilic, low fouling tendency for proteins)
Annealing	RO CA membranes annealed in hot water (70–80°C) to collapse skin macropores → higher salt rejection
pH stability	pH 4–8 only (hydrolysis of ester bonds outside this range) · Biological degradation by cellulase
Biomedical use	CuprophanTM (regenerated cellulose) → replaced by CA and CTA in modern hemodialyzers — low complement activation
Limitation	Biodegradable, narrow pH window, max T ~35–45°C, compaction under pressure

Membrane applications: Natural gas CO_2/CH_4 separation (commercial, acid gas plants) · Hemodialysis (CTA hollow fibers) · UF for protein separation · Microfiltration (depth filter) · Nano and ultrafiltration of sugars

5.8 Nafion® — Perfluorosulfonic Acid Polymer — Gold standard proton-exchange membrane (PEM)

Nafion is a sulfonated perfluoropolymer ($–CF_2–CF_2–$ backbone with $–OCF_2–CF(CF_3)–O–CF_2–CF_2–SO_3H$ side chains, EW = 900–1100 g/eq). The hydrophobic PTFE backbone provides chemical and thermal stability; the sulfonate-terminated side chains form hydrophilic ionic clusters (4–5 nm in size) that form a percolating proton transport network when hydrated. Proton conductivity ~ 0.1 S/cm at 80°C, 100% RH. The essential membrane in PEM hydrogen fuel cells (Toyota Mirai, Ballard stacks) and PEM electrolyzers for green hydrogen production.

Backbone	Perfluorocarbon (PTFE-like) + perfluoroether side chains terminating in $–SO_3H$ | EW: 900–1100 g/eq
Tg	Hydrated: ~110°C · Dry: ~225°C · Decomposes >280°C

Processing	Dissolve in EtOH/water (5–20 wt%) at 220°C/50 bar (autoclave) or commercial Nafion® dispersion (DuPont D521)
Film preparation	Cast from 5–15 wt% EtOH/water dispersion on glass → dry at 80°C → anneal at 140°C, 30 min
Ion conductivity	~0.1 S/cm (80°C, 100% RH) · drops dramatically below 60% RH — requires water management
H_2/O_2 crossover	<1 mA/cm² equivalent at 3 bar — critical for fuel cell safety (mixed potential, degradation)
Degradation	Chemical: radical attack by ·OH and ·OOH (generated at Pt cathode) → membrane thinning, pinhole formation
Alternatives	PFSA: Aquivion® (short side chain, higher Tg), Gore-Select® (ePTFE-reinforced, thinner 5–25 μm)
Vanadium RFB	Nafion N117 (183 μm) standard membrane; vanadium crossover and H_2O osmosis are key issues

Membrane applications: PEM hydrogen fuel cells · PEM water electrolyzers (green H_2) · Chlor-alkali electrolysis (Cl_2/NaOH production) · Vanadium redox flow batteries · CO_2 electroreduction reactors · Electrodialysis

5.9 PEBA — Poly(ether-b-amide) — Block copolymer elastomer · CO_2-selective and water vapor permeable

PEBA (Pebax®, Arkema) is a thermoplastic elastomer composed of rigid polyamide (PA6, PA12, or PA11) hard blocks and soft poly(ethylene oxide) (PEO) or poly(tetramethylene oxide) (PTMO) soft blocks. The PEO phase provides high CO_2 affinity and water vapor permeability. CO_2/N_2 selectivity 30–50 with CO_2 permeability 100–600 Barrers — significantly above the Robeson bound for many grades (Pebax 1074 is the benchmark grade). Easily processed from solution (EtOH/water or n-BuOH) into dense films or as a selective coating on porous hollow fibers.

Structure	–[PA hard block]–[PEO or PTMO soft block]– block copolymer | PA6/PEO (Pebax 1074): 40/60 wt%
Tg of soft block	PEO block: Tg ~ −60°C · PA hard block Tm: ~150–200°C · Two-phase microphase separated
Primary solvents	EtOH/water (70:30 v/v, reflux 78°C) · n-BuOH (reflux 118°C) · 1-propanol · EtOH alone (lower conc.)
Typical dope	Pebax 1074 5–8 wt% in EtOH/H_2O (70:30) at 80°C · No additional additives typically needed

Film preparation	Pour hot solution onto PTFE mold or PAN/PES support · slow evaporation at 40°C, 48 h · or spray-coat
CO_2 permeability	Pebax 1074: ~350 Barrers CO_2 · CO_2/N_2 ~50 · H_2O/N_2 >1000 (highly water vapor permeable)
CO_2 selectivity basis	PEO ether oxygens have high quadrupole interaction with CO_2 · $S(CO_2)/S(N_2)$ ~ 20–30
Modification	Blending with ionic liquids (1-butyl-3-methylimidazolium tetrafluoroborate) → further enhanced CO_2/N_2
Limitation	Requires careful film deposition (can be tacky); limited chemical resistance; some CO_2 plasticization at high pressure

Membrane applications: Post-combustion CO_2 capture (flue gas, TFC composite on hollow fiber) · Natural gas sweetening · Biogas upgrading · Water vapor recovery (dehumidification) · CO_2/H_2 separation in pre-combustion capture

5.10 Nylon 6,6 — Polyamide 6,6 — Step-growth polyamide · High mechanical strength · UF and hollow fiber applications

Nylon 6,6 (poly(hexamethylene adipamide)) is produced by step-growth condensation of hexamethylenediamine (HMD) and adipic acid, both produced petrochemically. It is a semi-crystalline polyamide with Tm 262°C and excellent tensile strength (>80 MPa). As a membrane material, Nylon 66 is used primarily in MF membranes (0.1–5 µm pore size) for analytical and sterilization filtration (Whatman, Pall, Millipore). Not widely used in gas separation due to low CO_2 permeability and competition from PES, PVDF, and PI.

Backbone	$–[NH–(CH_2)_6–NH–CO–(CH_2)_4–CO]_n–$ | MW: 20–60 kDa (membrane grade)
Tg / Tm	Tg: ~50–65°C (dry), ~20°C (wet, plasticized by water) · Tm: 262°C · Crystallinity: 35–50%
Primary solvents	Formic acid (85–100%, RT) · Trifluoroethanol · Phenol/formic acid mixtures · m-Cresol (hot)
TIPS diluents	Ethylene glycol, glycerol, and other high-boiling diols for TIPS processing
Typical dope (MF)	Nylon 66 12–18 wt% in formic acid · cast and immersed in water/EtOH bath
Membrane applications	MF membranes 0.1–5 µm · Analytical filter discs · Sterilization filtration · Solvent filtration

Contact angle	~50–70° (hydrophilic due to amide groups; absorbs ~8% water at saturation)
Mechanical strength	Tensile: 80 MPa dry · 60 MPa wet · Excellent burst pressure for hollow fiber modules
Limitation	Poor acid resistance (hydrolysis in strong acids) · Soluble in formic acid — limits cleaning options · Low CO_2 permeability

Membrane applications: MF analytical filtration (sterilization, particle removal) · Hollow fiber modules for bioprocess MF · Solvent filtration (organic-compatible grade) · Nanofiber membranes by electrospinning

5.11 Polyethylene / Polypropylene (PE / PP) — Polyolefin semi-crystalline · Microporous membranes by stretching

PE and PP microporous membranes are not made by NIPS but by controlled melt-extrusion followed by uniaxial stretching (the Celgard process) or by TIPS. In the stretching process: (1) extrude a lamellar semi-crystalline film at high draw-down ratio → oriented lamellar morphology; (2) cold-stretch to create crazes/microcracks between lamellae (incipient pores); (3) hot-stretch to open and stabilize pores (30–200 nm slit-shaped). Extremely low cost, chemically inert (resistant to pH 1–14, most solvents), and processable as battery separators and gas-liquid contactors.

Backbone	PP: $–(CH_2–CH(CH_3))_n–$ | PE: $–(CH_2–CH_2)_n–$ | Both semi-crystalline, high MW (>100 kDa)
Tm	PP: 160–170°C · HDPE: 130–135°C · LLDPE: 115–125°C
Processing	Melt extrusion + uniaxial stretching (Celgard process) · or TIPS in mineral oil/dioctyl phthalate
Pore structure	Slit-shaped pores 30–200 nm · tortuosity ~2–5 · porosity 35–55% · symmetric through thickness
Contact angle	Very hydrophobic: PP ~104°, PE ~95° · surface treatment (plasma, sulfonation) needed for water wetting
Key application	Li-ion battery separator (Celgard® 2400, 2500): 25 µm thick · PP or PP/PE/PP trilayer for thermal shutdown
Gas-liquid contactors	PP hollow fibers (Membrana X-40): CO_2 absorption, ozonation, degassing — exploit hydrophobicity
TIPS diluent	Mineral oil, dibutyl phthalate (DBP), dioctyl phthalate (DOP) for TIPS of PP at 200–220°C
Limitation	No solution casting (insoluble at RT) · Cannot make asymmetric structure by stretching · Requires surface activation for water filtration

Membrane applications: Li-ion battery separator · CO_2 gas-liquid contactors (PP hollow fiber) · Blood oxygenators (PP microporous hollow fiber) · Membrane distillation (MD) for desalination · Degassing of boiler feed water

5.12 Sulfonated Polyethersulfone / Sulfonated PSf (SPES / SPSf) — Charged membrane for nanofiltration and electromembrane applications

Sulfonation of PES or PSf backbone (–SO_3H groups grafted onto aryl ring) imparts cation exchange character and dramatically increased hydrophilicity. Degree of sulfonation (DS) controls IEC (ion exchange capacity), water uptake, and swelling. At DS ~ 1.0 (one –SO_3H per repeat unit), water uptake can exceed 100 wt% and membranes behave as hydrogels. Moderate sulfonation (DS 0.2–0.5) gives anti-fouling, charged NF membranes with Donnan exclusion of multivalent ions. Basis for many NF membranes used in dye removal and pharma.

Synthesis	PSf/PES + $ClSO_3H$ in 1,2-dichloroethane (0°C) · or SO_3/trimethyl phosphate · DS controlled by temp/time
Primary solvents	NMP (15–22 wt%) · DMAc · NMP/water · DMSO at elevated temperature
IEC	Typical IEC: 0.5–2.5 meq/g · Higher IEC → higher H^+ conductivity but excessive swelling
Water uptake	DS 0.3: ~15–30 wt% · DS 0.7: ~80–150 wt% · DS 1.0: >200 wt% (hydrogel-like)
Contact angle	DS 0.3: ~30–50° · DS 0.7: <10° · Superhydrophilic at high DS
NF performance	$MgSO_4$ rejection: 80–95% · NaCl rejection: 20–60% · MWCO 150–1000 Da · typical NF charge membranes
Stability	Limited long-term stability in alkali (sulfonation can cleave) · good acid stability · radiation stable

Membrane applications: Charged NF for dye wastewater treatment · Heavy metal ion removal (multivalent rejection) · Pharmaceutical separation · Proton exchange membrane (moderate-T fuel cells) · Electrodialysis (EDR)

6. DOPE FORMULATION — PRACTICAL GUIDE

A membrane dope (casting solution or spinning dope) must satisfy simultaneous requirements: sufficient polymer concentration for mechanical integrity, appropriate viscosity for the spinning/casting process, and a thermodynamic composition that places the system in the homogeneous region close enough to the binodal to enable rapid phase separation upon non-solvent contact.

6.1 Dope Components and Their Functions

Component	Role	Typical Examples	Effect on Morphology
Polymer	Structural matrix — forms membrane wall	PVDF, PES, PAN, PI, CA	Higher conc → denser structure, smaller pores, higher rejection
Primary solvent	Dissolves polymer → forms homogeneous dope	NMP, DMAc, DMF, DMSO, $CHCl_3$	Solvent quality affects binodal curvature and diffusion kinetics
Co-solvent	Lowers viscosity, adjusts binodal position	Acetone, THF, EtOH, dioxane	Can accelerate or decelerate exchange with non-solvent bath
Pore former (labile additive)	Phase-separates from polymer → creates pores	PVP, PEG, LiCl, glycerol, H_2O	Larger pore former MW → larger pores; higher loading → higher porosity
Macrovoid suppressor	Retards non-solvent ingress front	PVP K90, high-MW PEG, glycerol	Slows diffusion → sponge-like morphology; higher rejection
Inorganic additive	Surface modification or mixed-matrix	SiO_2, TiO_2, ZIF-8, Al_2O_3 NPs	Enhanced hydrophilicity (TiO_2); improved CO_2 transport (ZIF-8 MMM)
Non-solvent (in dope)	Moves dope composition closer to binodal	H_2O (1–5%), EtOH	Faster phase separation → finger-like + thinner skin

6.2 Comprehensive Dope Formulations by Polymer

Polymer	Formulation (wt%)	Viscosity (cP, 25°C)	Bore Fluid	Bath	Target Membrane

PVDF (NIPS, UF)	PVDF 18 / NMP 76 / LiCl 3 / H_2O 3	8,000–15,000	NMP/H_2O 50:50	H_2O 25°C	UF 10–100 kDa MWCO
PVDF (TIPS, MF)	PVDF 20 / TEG 75 / SiO_2 5	High melt viscosity	N/A (melt)	Cold H_2O 4°C	MF 0.1–0.5 µm
PVDF (HFMC, anti-wet)	PVDF 18 / NMP 74 / SiO_2-HDTMS 5 / LiCl 3	12,000–20,000	Water	H_2O 25°C	Anti-wetting HFMC fiber
PAN (UF hollow fiber)	PAN 14 / DMF 83 / LiCl 3	3,000–8,000	DMF/H_2O 70:30	H_2O 40°C	UF 10–50 kDa MWCO
PES (UF)	PES 20 / NMP 76 / PVP K30 4	10,000–25,000	NMP/H_2O 60:40	H_2O 20°C	UF 10–100 kDa MWCO
PSf (TFC substrate)	PSf 18 / NMP 78 / PEG 400 4	8,000–18,000	NMP/H_2O 50:50	H_2O 20°C	PSf UF support for TFC
PI — Matrimid (dense film)	Matrimid 25 / NMP 75	20,000–60,000	N/A (flat sheet)	Evaporation / MeOH	Dense gas separation film
CA (gas sep)	CA 22 / acetone 66 / formamide 10 / $MgClO_4$ 2	5,000–12,000	Isopropanol/H_2O	H_2O/IPA 5°C	Asymmetric gas sep CA membrane
PIM-1 (dense film)	PIM-1 6 / $CHCl_3$ 94	200–800	N/A (flat sheet)	Slow $CHCl_3$ evaporation, 24–48 h	Dense freestanding film for gas testing
PEBA (selective TFC layer)	Pebax 1074 6 / EtOH/H_2O 70:30	1,000–3,000	N/A (TFC coating)	Dip-coat on PAN or PES UF substrate	CO_2-selective composite membrane

Cellulose Acetate (RO)	CA 20 / DMAc 77 / LiCl 3	8,000–20,000	IPA/H_2O 60:40	H_2O 0–5°C + anneal 70°C	Asymmetric RO membrane

6.3 Critical Dope Preparation Steps

The dope preparation sequence is not trivial — incorrect steps cause polymer aggregation, undissolved gel particles, and defective membranes:

1. Weigh polymer and place in sealed amber vial with a subset (~60%) of solvent. Cap and place on roller mixer at 40–60°C for 12–24 h to swell polymer granules.
2. Add remaining solvent and any liquid additives (PVP solution, LiCl solution, plasticizer). Continue mixing 24–48 h at temperature until fully dissolved and optically clear.
3. Add inorganic additives (SiO_2 NPs, ZIF-8) last, sonicate dope 30 min at 40°C, then return to roller to prevent NP aggregation.
4. Degas under gentle vacuum (200–400 mbar) for 2–4 h at 25–40°C, or let stand 24 h to allow air bubbles to rise — bubbles cause pinholes in hollow fiber.
5. Filter through 20–50 μm stainless steel mesh to remove gel particles and undissolved aggregates.
6. Allow dope to equilibrate to spinning temperature (25–60°C) before extrusion. Monitor viscosity before each spin run.

CRITICAL

For hollow fiber spinning, bore fluid composition is as critical as the dope. A bore fluid of NMP/water (30–70% NMP) controls inner surface formation: more NMP → slower inner skin precipitation → more open inner surface (lower resistance). Water bore → fast inner precipitation → denser, tighter inner surface. For HFMC CO_2 contactors, a more open inner surface is preferred to allow rapid CO_2 transfer.

7. PERFORMANCE BENCHMARKING — THE ROBESON UPPER BOUND

The Robeson upper bound (1991, revised 2008) is the empirical limit of simultaneously achievable permeability and selectivity for dense polymeric membranes. It arises from the fundamental trade-off embedded in the solution-diffusion mechanism: polymers with higher free volume (higher permeability) typically have less size-discriminating chain packing (lower diffusivity selectivity).

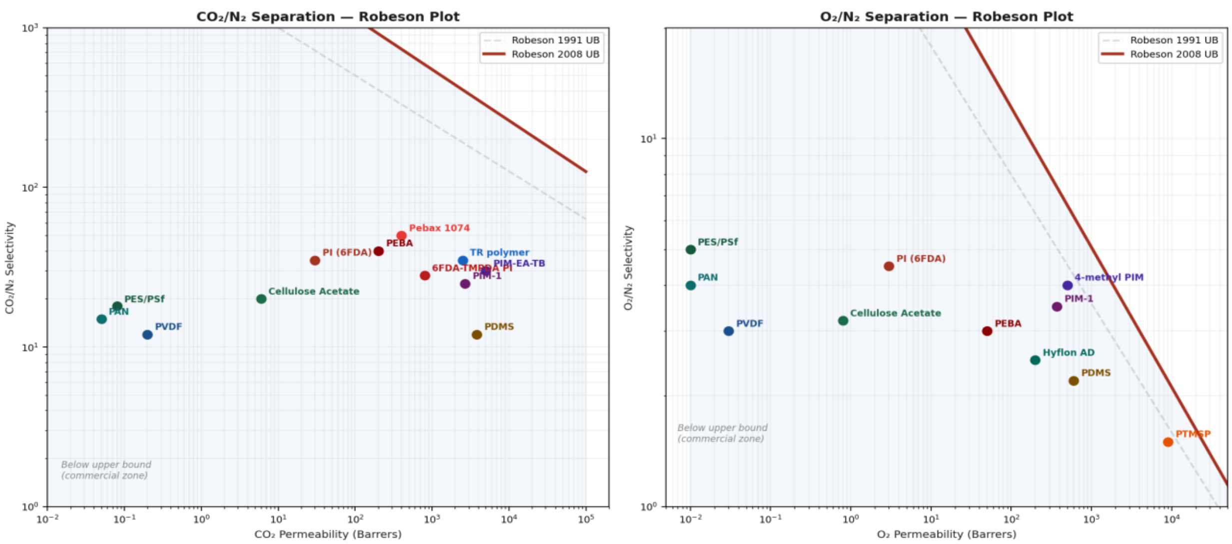

Figure 5 — Robeson upper bound plots for CO_2/N_2 and O_2/N_2 separation (key polymeric membrane materials)

Figure 6 — Robeson upper bound plots for CO_2/N_2 and O_2/N_2 separation with key polymeric membrane materials positioned

7.1 Reading the Robeson Plot

The Robeson plot is a log-log plot of permeability P_A (x-axis, Barrers) versus selectivity α(A/B) = P_A/P_B (y-axis). The upper bound has the functional form: α = k · P_A^n, where n < 0 reflects the trade-off. Key zones:

- Below the bound: commercially viable region — most currently deployed membranes operate here
- Above the bound (Robeson 2008): 'super-upper bound' performance — achieved by PIMs, TR polymers, and COF-MMMs
- Left lower zone: glassy polymers (PI, PSf, PES) — high selectivity but low permeability
- Right lower zone: rubbery polymers (PDMS) — high permeability but low selectivity (solubility-driven)
- Upper right: goal of modern membrane research — combine high P with high α (PIMs, TRs, mixed-matrix)

7.2 Key Gas Pairs and Benchmark Polymers

Gas Pair	Application	Target α	Target P (Barrers)	Best Polymers (2024)	Key Challenge
CO_2/N_2	Post-combustion CO_2 capture	>30	>1000 CO_2	PIM-EA-TB, TR-PI, PEBA-1074, 6FDA-TMPDA	Physical aging,

					moisture stability
CO_2/CH_4	Natural gas sweetening, biogas	>40	>50 CO_2	Matrimid, 6FDA-DAM, CA, PEBA	CO_2 plasticization at >15 bar
O_2/N_2	O_2 enrichment (medical, oxy-fuel)	>4	>10 O_2	PIM-1, 6FDA-DDBT, PTMSP, facilitated	Aging; cost vs. pressure swing adsorption
H_2/N_2	H_2 purification from N_2 streams	>50	>100 H_2	PIM-1 (H_2/N_2 ~100), PTMSP	Plasticization by impurities
H_2/CO_2	Pre-combustion H_2 capture	>20	>200 H_2	Stiff-chain PIs, TR polymers, porous carbon	H_2O and CO_2 compete for amine sites
H_2O/N_2	Air dehumidification	>>100	Very high H_2O	PEBA, PVA, crosslinked PEO	Condensation at high humidity

8. PVDF HOLLOW FIBER MEMBRANE CONTACTORS FOR CO_2 CAPTURE

Hollow fiber membrane contactors (HFMCs) combine the selectivity and high contact area of amine absorption with the compact geometry of hollow fiber modules. The membrane acts not as a selective barrier (both gas and liquid contact the membrane) but as a phase-contacting interface — preventing direct gas-liquid dispersion while maintaining high mass transfer rates.

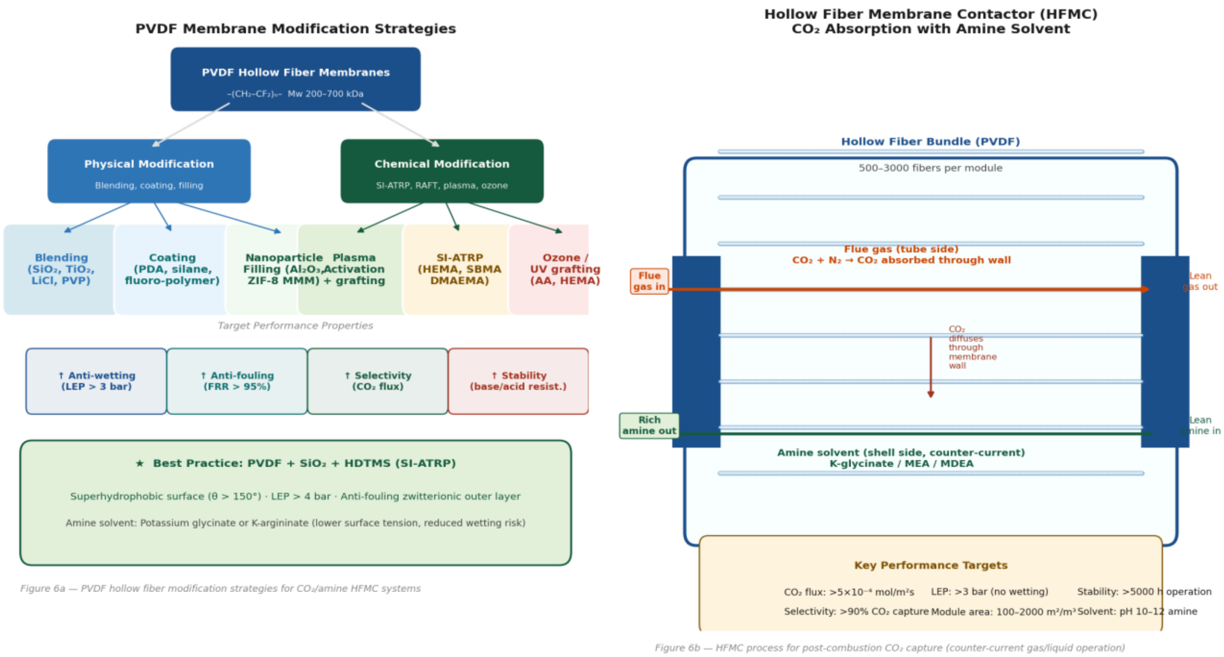

Figure 7 — PVDF hollow fiber modification strategies (left) and HFMC process schematic for CO_2/amine absorption (right)

8.1 Operating Principle

In a typical counter-current HFMC for post-combustion CO_2 capture:

7. Flue gas (CO_2 + N_2, 5–15% CO_2, 1 bar) flows through the lumen (tube side) of hollow PVDF fibers.
8. Amine solvent (MEA, MDEA, K-glycinate, K-argininate; 1–5 M, pH 10–13) flows counter-currently on the shell side.
9. CO_2 diffuses from the gas phase through the gas-filled membrane pores to the gas-liquid interface at the outer membrane surface (tube-side wetting) or inner surface (shell-side wetting mode).
10. At the interface, CO_2 reacts with the amine: $CO_2 + 2RNH_2 \rightarrow RNHCOO^- + RNH_3^+$ (carbamate for primary/secondary amines) or $CO_2 + H_2O + R_3N \rightarrow HCO_3^- + R_3NH^+$ (bicarbonate for tertiary amines).
11. Rich solvent is pumped to a thermal stripper (105–120°C) where CO_2 is released, amine regenerated, and returned — the absorption–stripping cycle.

8.2 Critical Performance Parameters for HFMC

Parameter	Definition / Formula	Target Value	How to Achieve

CO_2 flux (N_CO_2)	$N = K\_OL \cdot a \cdot \Delta C$ (mol/m^2s) — overall mass transfer rate	$>5\times10^{-4}$ mol/m^2s	Minimize membrane resistance (open porous structure, thin wall); high amine concentration
Liquid Entry Pressure (LEP)	Minimum ΔP to force liquid into pore: $LEP = -4\gamma\cos\theta/d\_p$	>3 bar (min) >5 bar (ideal)	Superhydrophobic surface ($\theta>150°$), small pore diameter (<0.2 μm); SiO_2-HDTMS coating
Overall mass transfer coeff (K_OL)	$1/K\_OL = 1/k\_g + 1/k\_m + 1/k\_L$ (gas, membrane, liquid resistances)	Maximize k_L (amine reaction)	Use reactive amine (MEA >> MDEA for rate); reduce membrane resistance by thin wall
Wetting ratio (x_wet)	Fraction of pores filled with liquid: $x\_wet = R\_m,wet/R\_m$	<5% (target <1%)	Superhydrophobic PVDF-SiO_2-HDTMS; K-glycinate (lower surface tension vs. MEA)
Module packing density	$a = 4\varepsilon/d\_o$ (m^2/m^3 module volume)	500–3000 m^2/m^3	Minimize fiber OD; maximize packing fraction ε (0.4–0.65 for random packing)
Long-term stability	LEP and CO_2 flux retention after >1000 h	>80% initial flux retention	Crosslinked or modified PVDF; SiO_2-HDTMS anti-wetting; amino acid salt amine
Amine compatibility	No membrane degradation in amine at pH 12, 60°C	No loss of LEP or tensile strength	Avoid ester-linked initiators (hydrolysis); use amide-bonded surface grafts; PVDF intrinsic base stability OK to pH 13

8.3 Amine Solvent Selection for PVDF HFMC

The amine solvent choice is as important as membrane design — it governs CO_2 absorption rate, energy of regeneration, and membrane wettability:

Solvent	Type	CO_2 Rxn Rate	Surface Tension (mN/m)	Regeneration Energy	Best For
MEA (30 wt%)	Primary amine	Very fast (k_2 ~6000 L/mol/s)	~55–62	High (Δh_regen ~4.2 GJ/tCO_2)	Benchmark — high capture rate, high energy
MDEA (40 wt%)	Tertiary amine	Slow (base-catalyzed only)	~47–50	Low (HCO_3^- product)	Low-pressure CO_2; blended with piperazine (PZ)

MEA + PZ (blended)	Primary + cyclic amine	Very fast	~52–58	Moderate	High CO_2 loading capacity; industrial standard
K-glycinate (3M)	Amino acid salt	Fast (k_2 ~3000 L/mol/s)	~65 (higher!)	Moderate	Non-volatile, thermally stable; lower aerosol loss; HFMC concern: higher surface tension → wetting risk
K-argininate (3M)	Amino acid salt (basic)	Fast	~60–64	Moderate–high	Natural amino acid, biodegradable, low toxicity; preferred for PVDF HFMC due to moderate surface tension
DETA (diethylenetriamine)	Polyamine	Fast	~50	High	High CO_2 capacity per kg; degradation concern
Ionic liquids ([bmim][BF_4])	IL + amine	Moderate	~35–42 (low!)	Very low (no vapor pressure)	Low wetting risk; very high cost limits scale-up

RECOMMENDATION (PVDF HFMC + CO_2)	For superhydrophobic PVDF-SiO_2-HDTMS hollow fibers: use K-argininate (3M, pH 11.5) as the absorption solvent. Reason: (1) amino acid salt — thermally stable, non-volatile, no aerosol formation; (2) surface tension ~60 mN/m — combined with LEP > 4 bar, wetting probability is minimal; (3) fast carbamate kinetics; (4) biodegradable. Pair with counter-current shell-side amine flow at 0.3–0.5 m/s linear velocity to maintain $k_L > 10^{-4}$ m/s.

9. SURFACE MODIFICATION OF POLYMERIC MEMBRANES

As-fabricated membranes rarely meet all performance requirements simultaneously. Surface modification tailors wettability, fouling resistance, ion selectivity, functional group density, and response to stimuli without changing the bulk mechanical properties of the membrane substrate.

9.1 Modification Strategy Comparison

Strategy	Mechanism	Key Chemistries	Durability	Membrane Applications
Plasma treatment	Radicals/ions from plasma activate surface, then react with O_2/NH_3/monomers	O_2 plasma → –OH, –COOH; NH_3 plasma → $–NH_2$; CF_4 plasma → superhydrophobic	Moderate (months–years); fades without crosslinking	Anti-fouling UF; hydrophilic PVDF for water wetting
UV/ozone grafting	UV photolysis of surface C–H or C–F bonds → radicals → graft monomer	PVDF + UV + AA/HEMA/DMAEMA → poly(AA) or poly(HEMA) grafts	Good (covalent grafts); UV degrades unfunctionalized surface if over-exposed	pH-responsive NF; anti-fouling UF; hydrophilization
SI-ATRP (surface-initiated)	Surface initiator + CRP → controlled brush growth from surface	PVDF–Br + CuBr/PMDETA + HEMA → PVDF-g-PHEMA brush; [Cu] ppm–100 ppm	Excellent (covalent C–C bonds); limited by ester initiator linkage stability in alkali	PVDF HFMC anti-wetting; anti-fouling UF; pH/T-responsive gates
SI-RAFT	Surface CTA + radical initiator → RAFT brush growth	PVDF–CTA + AIBN + DMAEMA → metal-free controlled brush; CS_2 group visible at 310 nm UV	Excellent; metal-free — preferred for biomedical, ion exchange membranes	Zwitterionic anti-fouling brushes; responsive gates; IEM modification
Polydopamine (PDA) coating	Oxidative self-polymerization of dopamine at pH 8.5, 24 h	PVDF/PES/PAN + 2 g/L dopamine pH 8.5 → PDA layer (5–50 nm) → secondary functionalization ($–NH_2$, –SH)	Moderate (physically adsorbed + some covalent quinone-amine bonds)	Universal anti-fouling primer; secondary functionalization of any membrane surface
Silane modification	Silanol (Si–OH) from silane hydrolysis condenses with surface –OH	PVDF + SiO_2–HDTMS → superhydrophobic; or APTES ($–NH_2$) for TFC adhesion	Good if condensed; poor if only physisorbed in humid environments	Superhydrophobic PVDF HFMC; TiO_2/SiO_2 anti-fouling layer coupling

Crosslinking	Thermal, UV, or chemical crosslinking of existing or grafted polymer	Glutaraldehyde crosslinking of PVA; UV crosslinking of CN groups in PIM-1; thermally rearranged PI	Excellent — reduces aging, improves chemical stability	Physical aging mitigation (PIM-1); PVA NF crosslinked; vitrimer membranes
Interfacial polymerization	MPD (water) + TMC (organic) → polyamide TFC selective layer	PA formed at PES/PSf support surface in 0.1–2% MPD + 0.05–0.5% TMC, hexane, 10–60 s	Excellent (covalent crosslinked PA layer)	RO/NF TFC membranes — highest commercial volume of any membrane type

9.2 Anti-Fouling Design Principles

Membrane fouling (deposition of proteins, cells, colloids, scale) is the primary operational challenge in water treatment and biomedical applications. Anti-fouling design strategies, from most to least effective:

12. Zwitterionic brushes (PSBMA, PCBMA): equal positive and negative charges → electrically neutral → zero protein adsorption (near-complete repulsion by hydration layer). The most effective anti-fouling surface known. Flux Recovery Ratio (FRR) >97% after BSA fouling in optimized systems.
13. PEGylated surfaces (PEO, PEG brushes): steric repulsion by hydrated PEG chains. Effective but oxidatively unstable at high O_2/H_2O_2 (degrades in bleach cleaning). FRR typically 85–95%.
14. Hydrophilic grafts (PHEMA, PVP, poly(acrylic acid)): moderate anti-fouling by hydration barrier. FRR 75–90%. Easier to prepare than zwitterionic but less durable.
15. Surface charge (negative, pH > 4): electrostatically repels negatively charged proteins and bacteria at neutral pH. Combined with hydrophilicity → synergistic anti-fouling. Common approach: sulfonation of PES.
16. Topographic structuring: micro/nanopillars (via imprinting) disrupt boundary layer flow and reduce contact area. Emerging, not yet industrial scale.

10. ADVANCED MEMBRANE CONCEPTS AND EMERGING FRONTIERS

The field of membrane science is experiencing a renaissance driven by demands for energy-efficient separations, decarbonization, and precision molecular separations. Several frontier concepts have moved from proof-of-concept to prototype stage within the last decade.

10.1 Mixed-Matrix Membranes (MMM)

MMMs embed inorganic or organic fillers (MOFs, zeolites, COFs, graphene oxide, carbon nanotubes) in a continuous polymer matrix. The goal is to combine the selectivity of the filler (molecular sieving) with the processability of the polymer. The Maxwell model predicts MMM permeability; but the real challenge is filler–polymer interface quality:

- Perfect interface: filler and polymer co-operate → performance exceeds both pure components
- Non-selective void (poor adhesion): gas bypasses filler → loss of selectivity
- Rigidified interfacial polymer layer: chains around filler immobilized → reduced flux
- Blocked filler pores: solvent or impurities clog MOF/zeolite channels during dope preparation

Leading MMM systems: ZIF-8/Matrimid (CO_2/CH_4), TpPa-1 COF/PVDF (CO_2/N_2), graphene oxide/PVA (H_2/CO_2), MIL-101/PDMS (vapor/gas), HKUST-1/6FDA-DAM.

10.2 Thermally Rearranged (TR) Polymers

Poly(hydroxyimide) or poly(hydroxybenzimidazole) precursor polyimides undergo solid-state thermal rearrangement at 350–450°C under inert atmosphere. The ring-opened backbone undergoes concerted rearrangement to a polybenzoxazole (PBO) or polybenzimidazole (PBI) — creating rigid, kinked structures with enlarged, hourglass-shaped free volume elements. CO_2 permeability increases 10–50× upon conversion; CO_2/N_2 selectivity can exceed 40 simultaneously. The high processing temperature limits substrates to inorganic or PI.

10.3 Facilitated Transport Membranes

Fixed-site or mobile carriers selectively bind and transport specific gases. Examples:

- CO_2: amine-functionalized membranes (polyvinylamine, polyallylamine) — CO_2 reacts reversibly with amine, forming carbamic acid, transported across, released on low-pressure side. CO_2/N_2 > 500 achieved at 110°C with humidified feed
- O_2: cobalt porphyrin or hemoglobin immobilized in polymer — O_2 forms reversible complex, desorbs at low pO_2
- Olefins: Ag^+-containing membranes ($AgBF_4$ in PVP) — selective π-complexation of ethylene or propylene over ethane/propane ($\alpha > 100$ at room temperature)
- Limitation: carrier saturation at high partial pressure; carrier degradation (oxidation, poisoning by H_2S)

10.4 Self-Healing and Adaptive Membranes

Dynamic covalent chemistry enables membranes that autonomously repair mechanical defects or adapt their permeability in response to environmental stimuli:

- Imine (Schiff base) crosslinked membranes: defects heal at room temperature by imine exchange in the presence of trace amine — relevant for long-term stability of hollow fiber CO_2 capture modules subjected to thermal cycling

- Disulfide vitrimer membranes: above topology-freezing temperature T_v, disulfide exchange enables flow and reprocessing — address irreversible fouling by allowing membrane recycling
- Responsive gates: poly(NIPAM) brushes on pore walls — pores open below LCST (32°C) and close above (coil-to-globule transition) — enable thermally gated permeability
- pH-responsive: poly(DMAEMA) or poly(acrylic acid) grafted pores — toggle between open (deprotonated) and closed (protonated and swollen) on pH change

10.5 Molecular Dynamics and Computational Membrane Design

Modern computational tools have transformed membrane material discovery:

- Molecular dynamics (MD) simulations: predict gas solubility, diffusivity, and chain mobility in specific polymers. Key software: GROMACS, LAMMPS, NAMD with DREIDING or PCFF force fields
- Grand Canonical Monte Carlo (GCMC): compute gas adsorption isotherms in MOFs and COFs — essential for MMM filler screening
- Transition-state theory (TST): predict activation energy barriers for gas diffusion jumps between polymer chain segments — rationalizes diffusivity selectivity
- Machine learning (ML) permeability prediction: neural networks trained on Polymer Genome database (>700 polymers) can predict P and α for new polymers from SMILES strings with $R^2 > 0.85$
- High-throughput virtual screening: screen 10^5–10^6 MOF structures for CO_2/N_2 separation with GCMC in 24 hours — then synthesize the top 10–20 candidates experimentally

11. MEMBRANE CHARACTERIZATION METHODS

Membrane characterization bridges synthesis and performance — without it, you cannot determine whether a modification has worked or understand why a membrane fails. The following techniques are the core toolkit.

Technique	What It Measures	Key Parameters / Information	Typical Use in Membrane Science
SEM (Scanning Electron Microscopy)	Surface and cross-section morphology	Skin thickness, macrovoid structure, pore size distribution (qualitative), filler dispersion in MMM	Standard first characterization after spinning; fracture cross-sections in liquid N_2
AFM (Atomic Force Microscopy)	Surface topography and roughness	Ra, Rq roughness; pore size and distribution; adhesion force mapping; Young's modulus (PeakForce QNM)	Skin layer pore size; anti-fouling surface characterization; protein adhesion mapping
FTIR-ATR (Infrared spectroscopy)	Chemical functional groups at surface (ATR: ~1 μm depth)	Identifies polymer backbone, grafted groups (C=O, O–H, N–H, C–F, C–N, SO_3^-); confirms surface modification success	Confirm SI-ATRP brush growth (HEMA carbonyl ~1730 cm^{-1}); PDA coating (catechol OH ~3300 cm^{-1})
XPS (X-ray Photoelectron Spectroscopy)	Elemental composition and bonding state at surface (0–10 nm depth)	F%, N%, C%, S%, O%: quantify grafting density; identify bond types (C–F, C–OH, C=O, C–N)	Confirm degree of SI-ATRP functionalization; measure sulfonation degree; detect filler element
Contact Angle (Water, oil)	Surface wettability — hydrophilicity / hydrophobicity	Advancing/receding angle; hysteresis; LEP (liquid entry pressure) for hollow fibers	Critical for HFMC: $\theta > 130°$ required for non-wetting; measure after each modification step
GPC / SEC (Gel Permeation Chromatography)	Molecular weight and distribution	Mn, Mw, PDI of bulk polymer or sacrificial free chains from SI-CRP dope	Determine dope polymer MW; track MW in SI-ATRP experiments via sacrificial initiator strategy
TGA (Thermogravimetric Analysis)	Thermal stability and composition	Onset decomposition temp, mass loss steps, inorganic filler content (residue at 600°C)	Filler loading in MMM; thermal stability of modified PVDF; PVP additive residual content

DSC (Differential Scanning Calorimetry)	Tg, Tm, crystallinity, phase transitions	ΔHm → crystallinity %; Tg of amorphous phase; multi-block copolymer transitions (e.g., PEBA hard/soft blocks)	PVDF crystallinity after TIPS/NIPS (affects mechanical and permeability); phase behavior in blends
BET (N_2 adsorption)	Surface area and pore size distribution	BET surface area (m^2/g), micropore volume, pore size by DFT analysis	Essential for PIM-1, MOF, COF, CMS membranes; MMM filler quality control
Gas Permeation Testing	Direct measurement of P and α	Permeability P_i (Barrers), diffusivity D, solubility S, ideal selectivity $\alpha(A/B) = P_A/P_B$	Primary performance metric for gas separation membranes; use constant-pressure or time-lag method
Liquid Permeation / Rejection	Flux and solute rejection	Pure water flux ($L/m^2h/bar$), solute rejection R% at specified MWCO probe (dextrans, PEG), MWCO	UF/NF characterization; anti-fouling testing (BSA flux recovery ratio FRR)
WAXD / SAXS	Crystal structure and d-spacing	Lamellar spacing, crystalline phase identification (PVDF α/β/γ), interlayer spacing in 2D materials	PVDF phase identification; MOF/COF crystallinity in MMM; track physical aging in PI (d-spacing shift)
Liquid Entry Pressure (LEP)	Minimum pressure to wet hydrophobic membrane	$LEP = -4\gamma\cos\theta/d_p$ — determines non-wetting operating window	Critical for HFMC: target LEP > 3 bar for CO_2/amine systems; measure as function of amine type and conc.

2. Key Equations and Design Rules — Quick Reference

Equation / Rule	Formula	Variables / Notes
Permeability	$P = J \cdot l / \Delta p = S \cdot D$	P in Barrers (1 Barrer = 10^{-10} $cm^3(STP)\cdot cm / cm^2\cdot s\cdot cmHg$) | S = solubility | D = diffusivity
Ideal selectivity	$\alpha(A/B) = P_A / P_B = (S_A/S_B)(D_A/D_B)$	Diffusivity selectivity often dominates in glassy polymers; solubility selectivity in rubbery
Carothers equation	$\bar{X}n = 1 / (1 - p)$	p = fractional monomer conversion | $\bar{X}n$ = number-average degree of polymerization

Flory-Huggins theory	$\Delta G_{mix} / nRT = \varphi_1 \ln(\varphi_1) + \varphi_2/r \cdot \ln(\varphi_2) + \chi_{12}\varphi_1\varphi_2$	χ_{12} = Flory-Huggins interaction parameter (< 0.5 for miscible) | r = degree of polymerization
Liquid Entry Pressure	$LEP = -4B\gamma\cos\theta / d_{max}$	B = geometric factor (~1 for cylindrical pore) | γ = surface tension (N/m) | θ = contact angle | d_max = largest pore diameter
Hagen-Poiseuille (UF flux)	$J = \varepsilon \cdot d_p^2 \cdot \Delta P / (32 \cdot \mu \cdot l \cdot \tau)$	ε = porosity | d_p = pore diameter | μ = viscosity | τ = tortuosity | l = membrane thickness
Knudsen selectivity	$\alpha_K(A/B) = \sqrt{M_B / M_A}$	M = molecular weight (g/mol) | Applies when $d_p < \lambda$ (mean free path ~66 nm at 1 atm for N_2)
Overall mass transfer (HFMC)	$1/K_{OL} = 1/k_g + 1/k_m + m/k_L$	k_g, k_m, k_L = gas, membrane, liquid mass transfer coefficients | m = solubility/distribution coefficient
Time-lag diffusivity	$D = l^2 / 6\theta$	θ = time-lag (x-axis intercept of steady-state flux vs. time plot) | l = membrane thickness
Robeson upper bound	$\log \alpha = n \cdot \log P + \log k$	n and k are gas-pair specific constants from Robeson (2008) | n ~ −0.35 for CO_2/N_2
Mark-Houwink-Sakurada	$[\eta] = K \cdot M^a$	[η] = intrinsic viscosity | K and a = polymer/solvent specific constants (from GPC standards)
Free volume (Cohen-Turnbull)	$D = A \cdot \exp(-B / V_f)$	Vf = fractional free volume = $(V - V_0) / V$ | B = overlap factor × critical free volume for diffusion jump
Concentration polarization	$C_m / C_b = \exp(J / k)$	C_m = membrane surface conc | C_b = bulk conc | k = mass transfer coefficient (m/s) | J = flux (m/s)
Donnan exclusion (IEM)	$c_{\pm,m} = c_\pm / (1 + (c_{fix}/2c_\pm)^2)^{0.5}$	c_fix = fixed charge concentration in membrane | $c_\pm$ = electrolyte concentration in solution

13. INDUSTRIAL AND COMMERCIAL MEMBRANE EXAMPLES

The following table compiles specific commercial membrane products, their manufacturers, key specifications, and the application context. These are real-world benchmarks against which laboratory membranes are evaluated.

Product / Module	Manufacturer	Polymer	Geometry	Application	Key Spec
Celgard® 2400	Celgard LLC (USA)	Polypropylene (PP)	Flat sheet microporous	Li-ion battery separator	25 µm thick; 0.04×0.12 µm slit pores; Gurley: 620 s
Liqui-Cel® X-40	3M (Membrana)	Polypropylene (PP)	Hollow fiber (microPP)	Gas-liquid contactor (CO_2/O_2 removal)	10,000 m^2 contact area per module; OD 300 µm
FilmTec BW30-400	DuPont (former Dow)	Polyamide TFC / PSf	Spiral wound	Brackish water RO	Flux: 40 L/m^2h at 15.5 bar; NaCl rejection: 99.5%
FilmTec SW30-HR	DuPont	Polyamide TFC / PSf	Spiral wound	Seawater RO (SWRO)	Salt rejection: 99.75%; Flux 26 L/m^2h at 55 bar
NF270	DuPont	Semi-aromatic PA TFC	Spiral wound	Nanofiltration (divalent ion removal)	$MgSO_4$ rejection >97%; NaCl ~50%; MWCO 200–300 Da
Matrimid® hollow fiber	PRISM / Air Products	Matrimid® 5218 PI	Hollow fiber	O_2/N_2 enrichment; CO_2/CH_4 nat. gas	O_2/N_2 selectivity ~6; CO_2/CH_4 ~30
MEDAL™ modules	Air Liquide	PI / blended PI	Hollow fiber	H_2 recovery, CO_2 separation	H_2/CO selectivity

					~50; CO_2/CH_4 ~30
Polaris CO_2 module	MTR Inc. (USA)	PEBA + PEO composite	Spiral wound	CO_2 capture from flue gas / biogas	CO_2 permeance >1000 GPU; CO_2/N_2 ~50
Evonik SEPURAN Green	Evonik Industries	PDMS + PI composite	Hollow fiber	Biogas upgrading (CO_2/CH_4)	CH_4 purity >97%; >98% CH_4 recovery
Fresenius FX CorDiax	Fresenius Medical Care	Helixone® (PAES blend)	Hollow fiber	High-flux hemodialysis	β_2-microglobulin removal: >70%; UF coeff: 60 mL/h/mmHg
Gambro Polyflux 210H	Baxter International	Polyarylethersulfone + PVP	Hollow fiber	High-flux hemodialysis	High albumin-saving flux; middle molecule removal
Nafion™ N117	Chemours (DuPont spin-off)	Perfluorosulfonic acid	Flat sheet	PEM fuel cell, electrolyzer, vanadium RFB	Thickness: 183 µm; IEC: 0.91 meq/g; σ_H: ~0.1 S/cm (80°C)
DuraMem® 300	Evonik	Crosslinked PAN / polyimide	Flat sheet	Organic solvent NF (OSN)	MWCO 300 Da in ethanol; resistant to DMF, THF, acetone

14. CURRENT RESEARCH FRONTIERS AND OUTLOOK (2024–2030)

The following strategic areas represent the most active and promising directions in polymeric membrane science over the next 5–10 years, with direct relevance to CO_2 capture, water treatment, energy conversion, and precision molecular separations.

GRAND CHALLENGE	Energy-efficient CO_2 capture at scale: the IEA estimates that CCS must reach 7.6 Gt CO_2/year by 2050 to meet net-zero. Membrane-based capture (polymeric + hybrid) could reduce the energy penalty to <1.5 GJ/tonne CO_2 vs. 3.5–4.2 GJ/tonne for amine scrubbing, IF membrane permeance >1000 GPU and CO_2/N_2 selectivity >50 can be simultaneously achieved at pilot scale.

Frontier Area	Key Technical Goals (2025–2030)	Leading Approaches	Membrane Relevance
Post-combustion CO_2 capture	Membrane permeance >1000 GPU; CO_2/N_2 >50; stable >5000 h in humid flue gas	PEBA-1074 TFC; TR-polymer; facilitated transport (polyvinylamine); HFMC with amino acid salts	High: PVDF HFMC with K-argininate; composite PEBA on PAN substrate
Direct Air Capture (DAC)	CO_2 concentration at 420 ppm — requires extraordinary selectivity or temperature swing	Facilitated membranes + swing operation; liquid sorbent contactor (HFMC); electro-swing with redox carriers	Moderate: PP or PVDF HFMC as gas-liquid contactor stage before electrochemical DAC
Green hydrogen (PEM electrolyzer)	Membrane thickness <20 µm; H_2 crossover <1 mA/cm²; stable at 80°C, 60 bar; cost <\$20/m²	Reinforced Nafion (Gore-Select, Nafion XL); PFSA short-side-chain (Aquivion); hydrocarbon PEM (SPEEK, SPSU)	Very high: all PEM electrolyzers depend on proton-exchange membrane
Organic solvent nanofiltration (OSN)	Stable in DMF, THF, acetone, toluene; MWCO 100–500 Da; >30 L/m²h/bar	Crosslinked PI (P84, Matrimid + p-xylenediamine); PAN + crosslinking; PDMS TFC; ceramic backup	High: pharmaceutical synthesis, catalyst recovery, API concentration in organic solvents

Lithium recovery (DLE)	Li/Mg selectivity >1000; Li flux >0.5 mol/m²h; stable in brine pH 2–9	$Li_{1.33}Mn_{1.67}O_4$ / PVDF adsorptive membrane; ion-imprinted PVDF-SiO_2 membranes; LiSICON ceramic-polymer composite	High: emerging strategic importance for battery supply chain
Self-healing membranes	Autonomous crack repair in <1 h at ambient T; maintain LEP >2 bar after healing; no external trigger	Imine DCP network; disulfide vitrimer PDMS; microencapsulated amine/epoxide	High: long-term reliability of hollow fiber CO_2 capture modules; DCMD for seawater
Machine-learning polymer design	Predict P and α from molecular structure; close the simulation–synthesis–test loop in <2 weeks	Graph neural networks on Polymer Genome; GC-GCMC for MOF screening; active learning with robotic synthesis	Transformative: could identify next PIM-1 or PEBA breakthrough 100× faster than trial-and-error
Aquaporin biomimetic membranes	Water permeability 10× conventional RO; perfect NaCl rejection; scalable reconstitution	Aquaporin Z (AqpZ) in block copolymer bilayer; imidazole water channel (I-quartet); pillar[5]arene nanotubes	Research stage: Aquaporin A/S commercial pilot HF modules launched 2023

15. GLOSSARY OF KEY TERMS

Term	Definition
Asymmetric membrane	Membrane with a graded structure: dense skin on one face, porous sublayer below — both from the same polymer
Barrer	Unit of gas permeability: 1 Barrer = 10^{-10} cm³(STP)·cm / (cm²·s·cmHg)
Binodal	Curve in ternary phase diagram separating the one-phase and two-phase regions
Bore fluid	Fluid injected through the inner channel of a spinneret during hollow fiber spinning to form and support the inner lumen surface
CRP / CRP	Controlled radical polymerization — ATRP, RAFT, NMP — giving narrow PDI
Dead-end filtration	Filtration where all feed flows perpendicular to membrane; concentrate accumulates on membrane surface
Dope	Polymer solution used to fabricate membranes by phase inversion or electrospinning
Facilitated transport	Transport mechanism involving reversible chemical reaction of permeant with a carrier — enhances both flux and selectivity
Finger-like macrovoids	Large elongated pores formed during NIPS when solvent/non-solvent exchange is fast — run perpendicular to membrane surface
Free volume	Unoccupied volume in a polymer matrix — determines gas diffusivity and aging behavior of glassy membranes
FRR (Flux Recovery Ratio)	FRR (%) = $(J_2/J_0) \times 100$, where J_2 = flux after washing and J_0 = initial pure water flux — measures anti-fouling effectiveness
Glass transition temperature (Tg)	Temperature below which a polymer is glassy (rigid, low free volume) and above which it is rubbery (mobile chains, high free volume)
GPU (Gas Permeation Unit)	1 GPU = 10^{-6} cm³(STP) / (cm²·s·cmHg) — unit of membrane permeance (for thin films where thickness is not independently measured)
HFMC (Hollow Fiber Membrane Contactor)	Gas-liquid contactor using hollow fiber membrane as interface — membrane is not the selective barrier; phase contact occurs at membrane surface

IEC (Ion Exchange Capacity)	Milliequivalents of exchangeable ions per gram of dry membrane — quantifies charged group density in IEM
LEP (Liquid Entry Pressure)	Minimum transmembrane pressure needed to force a liquid to enter the pores of a hydrophobic membrane — key metric for HFMC anti-wetting performance
Living polymerization	Polymerization without irreversible chain transfer or termination — chains remain active indefinitely, enabling block copolymer synthesis
Maxwell model (MMM)	$P_{eff} = P_m \cdot (P_f + 2P_m - 2\varphi_f(P_m - P_f)) / (P_f + 2P_m + \varphi_f(P_m - P_f))$ — predicts permeability of mixed-matrix membrane from filler and matrix properties
MWCO (Molecular Weight Cutoff)	Molecular weight at which a UF membrane achieves 90% rejection — used to characterize pore size indirectly
NIPS	Non-Solvent Induced Phase Separation — the dominant membrane fabrication method; immersion of dope in non-solvent bath
PDI (Polydispersity Index)	PDI = Mw/Mn — ratio of weight-average to number-average molecular weight; 1.0 = perfectly monodisperse; >2 = broad distribution
Permeance	P/l — permeability normalized by membrane thickness (GPU) — the practically measurable quantity for composite or hollow fiber membranes
Physical aging	Gradual densification of glassy polymer free volume over time — reduces gas permeability; primary long-term stability issue for PIM-1 and PI
Plasticization	Absorption of condensable gas (CO_2, H_2S) into glassy polymer → chains mobilized → enhanced diffusivity but loss of selectivity — critical for high-pressure CO_2 service
Robeson upper bound	Empirical limit of simultaneously achievable permeability and selectivity for dense polymeric membranes, first published 1991, updated 2008
SI-ATRP	Surface-Initiated ATRP — controlled radical polymerization of monomer from initiators covalently anchored to a solid substrate surface
Spinodal decomposition	Spontaneous phase separation by concentration fluctuations that grow continuously — gives bicontinuous pore morphology inside spinodal region

TFC (Thin-Film Composite)	Membrane with an ultra-thin selective layer (polyamide, silicone) deposited on a porous support — highest performance RO/NF membranes
TIPS	Thermally Induced Phase Separation — phase separation driven by cooling a hot polymer/diluent solution; preferred for semi-crystalline PVDF, PP, PE
TR polymer (Thermally Rearranged)	Polyimide precursor converted by solid-state thermal rearrangement (350–450°C) to polybenzoxazole — dramatically enhanced gas permeability
Wetting (membrane)	Infiltration of liquid into hydrophobic membrane pores — destroys mass transfer driving force in HFMC and DCMD — catastrophic failure mode

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